5 Marks Questions

Q1. Explain OT (Operational Technology) and IT (Information Technology) networks, and differentiate between them.

OT Networks:

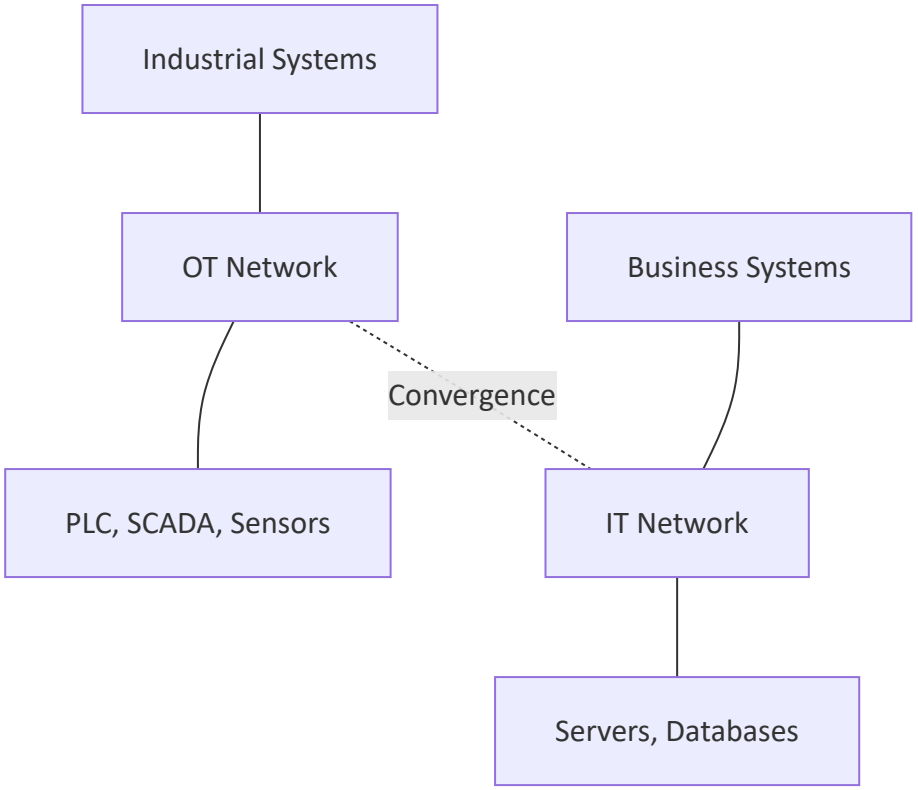
- Monitor and control physical industrial processes.
- Used in factories, power plants, utilities.
- Prioritize safety, reliability, and uptime.
- Include SCADA, PLC, DCS systems.
- Handle real-time deterministic communication.

IT Networks:

- Manage and process business information.
- Used in offices, data centers, enterprises.
- Prioritize confidentiality and data integrity.
- Include servers, databases, ERP systems.
- Handle best-effort communication.

Differentiation:

Factor	OT Networks	IT Networks
Purpose	Control physical processes	Manage business data
Priority	Safety and availability	Confidentiality and integrity
Latency	Real-time deterministic	Best-effort acceptable
Lifespan	15-20 years	3-5 years
Protocols	Modbus, Profibus, HART	TCP/IP, HTTP, SMTP
Updates	Rare and risky	Frequent and easy
Security	Physical isolation	Firewalls and encryption
Users	Engineers and operators	Employees and managers



Q3. Describe the physical design of IoT, including the various components and devices involved.

Physical Design:

- Refers to actual hardware components in IoT.
- Includes devices, sensors, and actuators.
- Encompasses networking and communication hardware.
- Defines physical topology of IoT systems.

Things/Devices:

- Sensors: collect data from environment.
- Actuators: perform actions on environment.
- Controllers: process data and control devices.
- Gateways: bridge devices to network.

Sensors:

- Temperature, humidity, pressure sensors.
- Motion, proximity, light sensors.
- Gas, smoke, water quality sensors.

Actuators:

- Motors, relays, solenoids.
- Smart locks and valves.
- LED lights and displays.

Controllers:

- Microcontrollers like Arduino, ESP32.

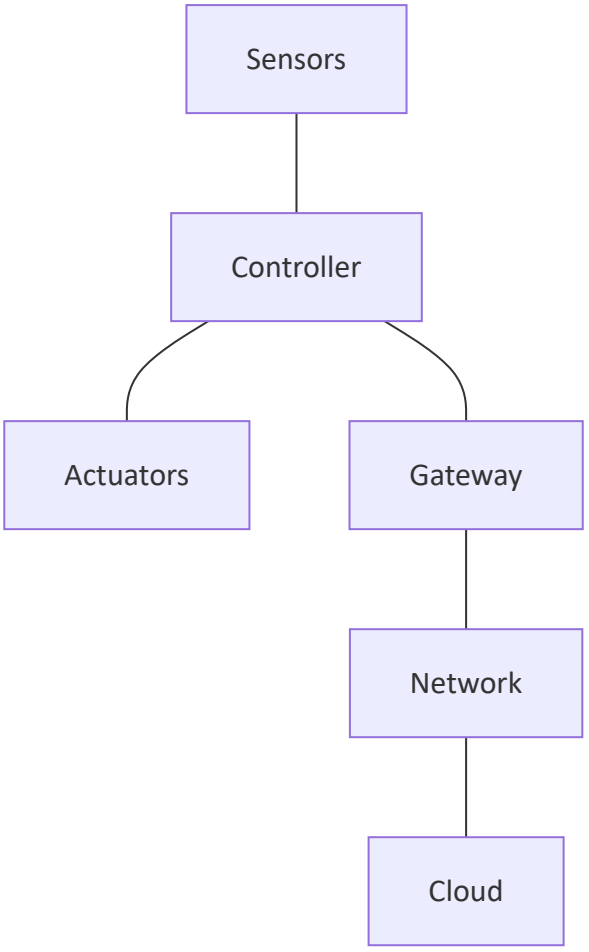
- Single-board computers like Raspberry Pi.
- PLCs for industrial control.

Gateways:

- Aggregate data from multiple devices.
- Protocol translation between networks.
- Edge processing capabilities.

Communication Hardware:

- Wi-Fi, Bluetooth, ZigBee modules.
- LoRa, NB-IoT, cellular modems.
- Ethernet and serial interfaces.



Q5. Explain the logical design of IoT, including its architectural layers and protocols.

Logical Design:

- Abstract representation of IoT system.
- Defines how components interact logically.
- Includes functional blocks and protocols.
- Independent of physical implementation.

Architectural Layers:

Perception Layer:

- Collects data from physical world.
- Includes sensors and actuators.
- Converts physical signals to digital data.

Network Layer:

- Transmits data between devices.
- Uses protocols like MQTT, CoAP.
- Handles routing and addressing.

Processing Layer:

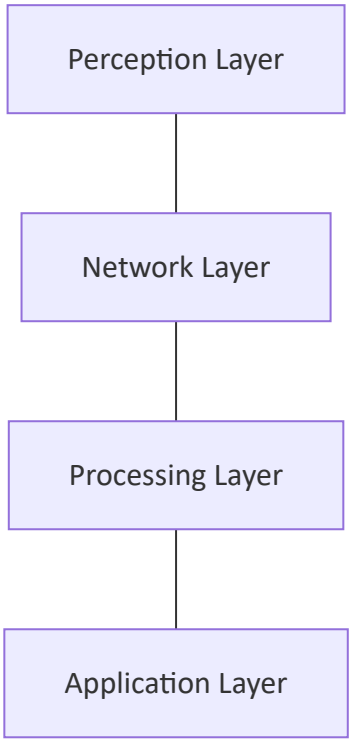
- Processes and analyzes data.
- Includes edge and cloud computing.
- Performs data filtering and aggregation.

Application Layer:

- Provides user-facing services.
- Includes dashboards and APIs.
- Delivers IoT applications to users.

Protocols:

- MQTT: lightweight publish-subscribe.
- CoAP: constrained application protocol.
- HTTP: request-response protocol.
- AMQP: advanced message queuing.
- XMPP: real-time messaging protocol.



Q6. Describe Bluetooth Low Energy technology, its architecture, and its significance in IoT communication.

Bluetooth Low Energy:

- Wireless personal area network technology.
- Designed for low power consumption.
- Operates in 2.4 GHz ISM band.
- Range typically 10-100 meters.
- Ideal for battery-powered IoT devices.

Architecture:

Controller:

- Physical layer and link layer.
- Handles radio communication.
- Manages advertising and scanning.

Host:

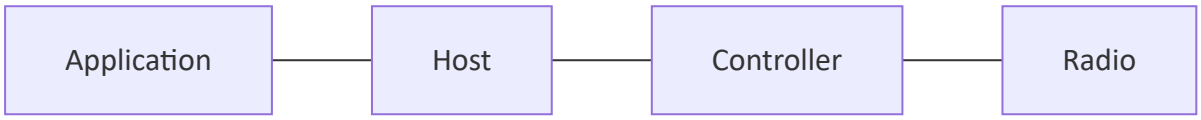
- Logical link control and adaptation.
- Attribute protocol (ATT).
- Generic attribute profile (GATT).
- Security manager protocol.

Application:

- User application profiles.
- GATT services and characteristics.
- Custom and standard profiles.

Significance in IoT:

- Ultra-low power consumption.
- Enables years of battery life.
- Low cost and small form factor.
- Widely supported in smartphones.
- Easy pairing and connection.
- Suitable for wearables and sensors.
- Supports mesh networking.



Q8. Explain NFC and RFID technologies, and compare them in terms of their working, range, and applications.

NFC:

- Near Field Communication technology.
- Operates at 13.56 MHz frequency.
- Range up to 10 cm.
- Enables two-way communication.

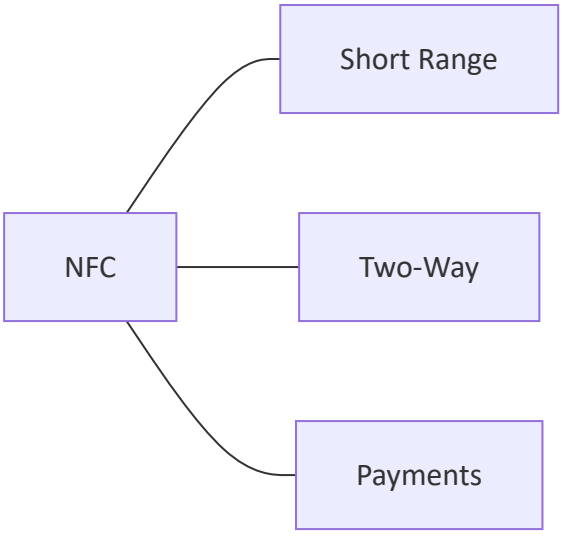
- Used for contactless payments.
- Supports peer-to-peer data transfer.

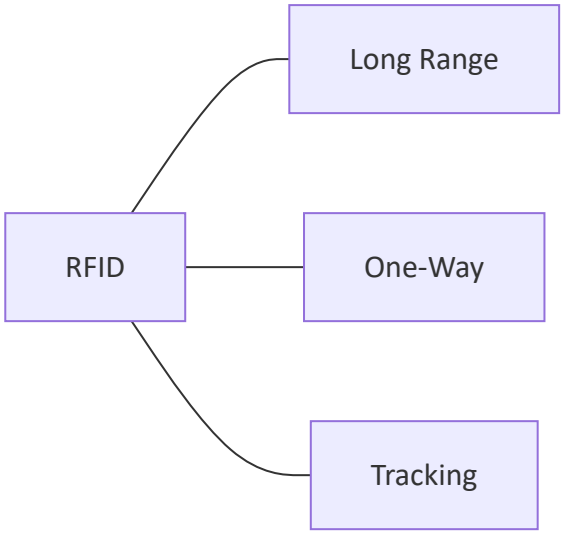
RFID:

- Radio Frequency Identification technology.
- Uses radio waves for identification.
- Range from few cm to several meters.
- Includes tag and reader components.
- Passive tags powered by reader.
- Active tags have own power source.

Comparison:

Factor	NFC	RFID
Frequency	13.56 MHz	125 kHz to 5.8 GHz
Range	Up to 10 cm	Up to 100 meters
Communication	Two-way	One-way mostly
Standard	ISO 14443, 18092	ISO 18000 series
Power	Active and passive	Passive and active
Cost	Higher	Lower
Applications	Payments, access	Inventory, tracking
Data Transfer	Yes	Limited





Q9. Differentiate between Edge, Fog, and Cloud computing, and establish their hierarchical relationship.

Edge Computing:

- Processing at data source.
- Lowest latency and bandwidth.
- Limited computational resources.
- Real-time decision making.
- Examples: smart sensors, PLCs.

Fog Computing:

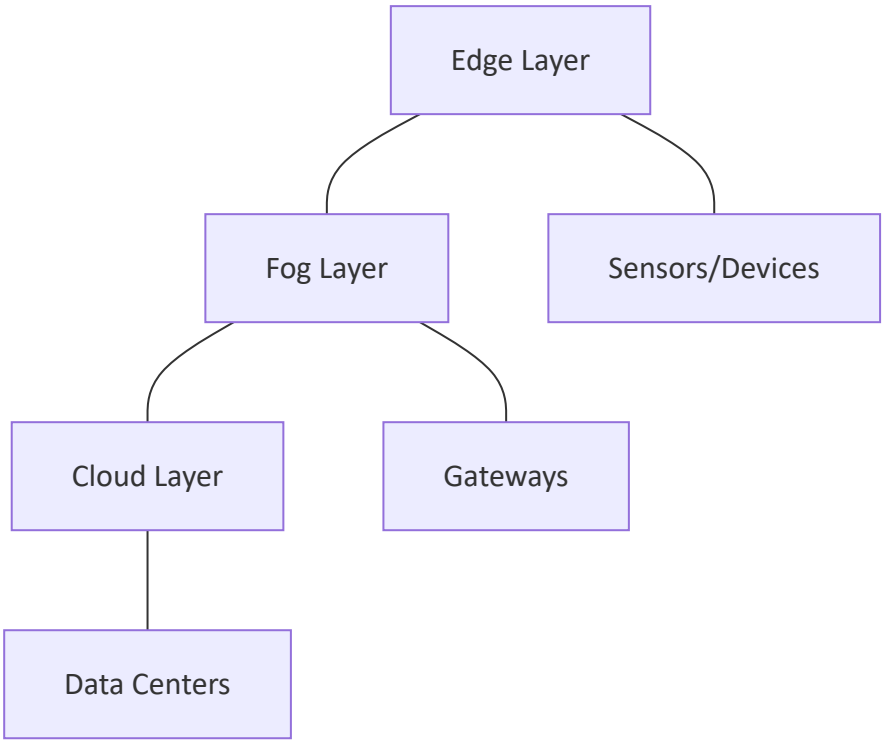
- Processing between edge and cloud.
- Intermediate layer of computing.
- Reduces data sent to cloud.
- Moderate latency and resources.
- Examples: gateways, local servers.

Cloud Computing:

- Centralized processing and storage.
- Highest latency and bandwidth.
- Unlimited computational resources.
- Long-term data storage.
- Examples: AWS, Azure, Google Cloud.

Hierarchical Relationship:

Factor	Edge	Fog	Cloud
Location	Data source	Local network	Remote data center
Latency	Very Low	Low	High
Bandwidth	Minimal	Moderate	High
Compute	Limited	Moderate	Unlimited
Storage	Very Less	Moderate	Massive
Cost	Low	Medium	High
Use Case	Real-time control	Local analytics	Big data analytics



Q10. With the help of a neat diagram, elaborate on the architecture of the IoTWF (IoT World Forum) reference model.

IoTWF Reference Model:

- Seven-layer architecture for IoT.
- Developed by IoT World Forum.
- Provides standardized framework.
- Addresses interoperability challenges.
- Covers edge to cloud continuum.

Layer 1 - Physical Devices:

- Sensors, actuators, and controllers.
- Physical things in IoT system.
- Data generation and action execution.

Layer 2 - Connectivity:

- Communication between devices.
- Protocols and network technologies.
- Wired and wireless connections.

Layer 3 - Edge Computing:

- Data processing at edge.
- Filtering and optimization.
- Protocol conversion and analytics.

Layer 4 - Data Accumulation:

- Data storage and aggregation.
- Converting data to information.
- Database and storage management.

Layer 5 - Data Abstraction:

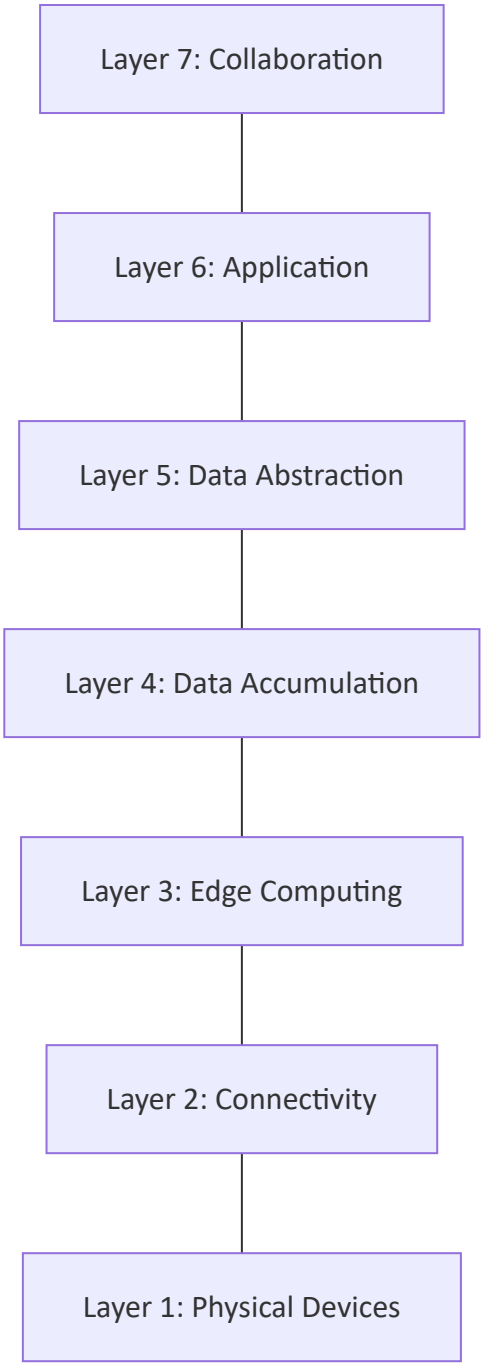
- Data normalization and integration.
- Creating unified data views.
- Managing diverse data sources.

Layer 6 - Application:

- Business applications and services.
- User interfaces and dashboards.
- Industry-specific solutions.

Layer 7 - Collaboration:

- People and process integration.
- Business workflow integration.
- Cross-organizational collaboration.



Q11. Explain the functional blocks of IoT and describe the role of each block.

Functional Blocks:

- Building blocks of IoT systems.
- Define core capabilities of IoT.
- Enable modular system design.
- Include identification, sensing, communication.
- Also include actuation, services, management.
- Security is cross-cutting concern.

Identification:

- Unique identity for each device.
- Uses UUID, URI, EPC standards.

- Enables addressing and tracking.

Sensing:

- Collects data from environment.
- Includes various sensor types.
- Converts physical to digital signals.

Communication:

- Transmits data between devices.
- Uses various protocols and networks.
- Enables device connectivity.

Actuation:

- Performs actions on environment.
- Uses actuators and controllers.
- Executes commands from system.

Services:

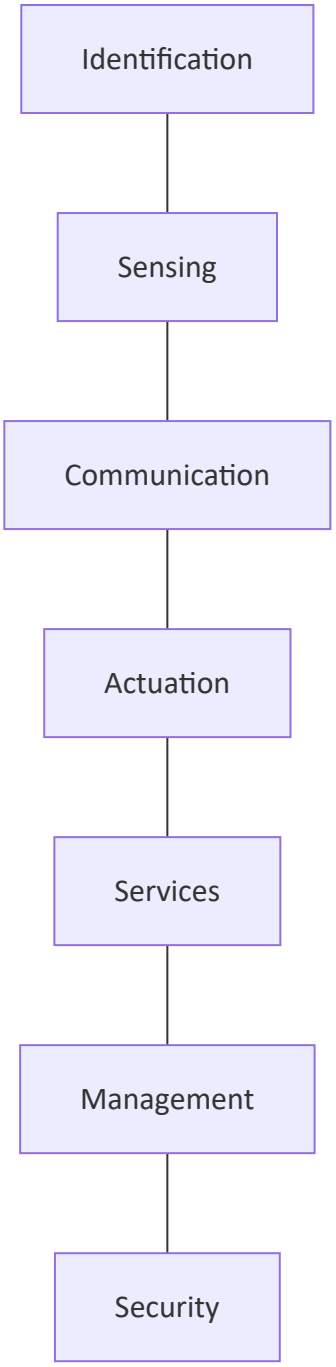
- Provides IoT functionalities.
- Includes data analytics, storage.
- Cloud and edge services.

Management:

- Device and network management.
- Configuration and monitoring.
- Firmware updates and diagnostics.

Security:

- Protects data and devices.
- Authentication and encryption.
- Access control and privacy.



Q12. Describe the various ZigBee devices and explain their operational modes and functions.

ZigBee:

- Low-power wireless mesh networking protocol.
- Based on IEEE 802.15.4 standard.
- Operates in 2.4 GHz, 900 MHz bands.
- Designed for low data rate applications.
- Supports large mesh networks.

ZigBee Devices:

Coordinator:

- Starts and manages the network.

- Selects channel and PAN ID.
- Allows devices to join network.
- Routes messages between devices.
- Only one per ZigBee network.

Router:

- Extends network coverage.
- Routes messages between devices.
- Always powered on device.
- Can have child devices.
- Mains-powered typically.

End Device:

- Performs specific sensing tasks.
- Can sleep to save power.
- Cannot route messages.
- Battery-powered typically.
- Communicates with parent.

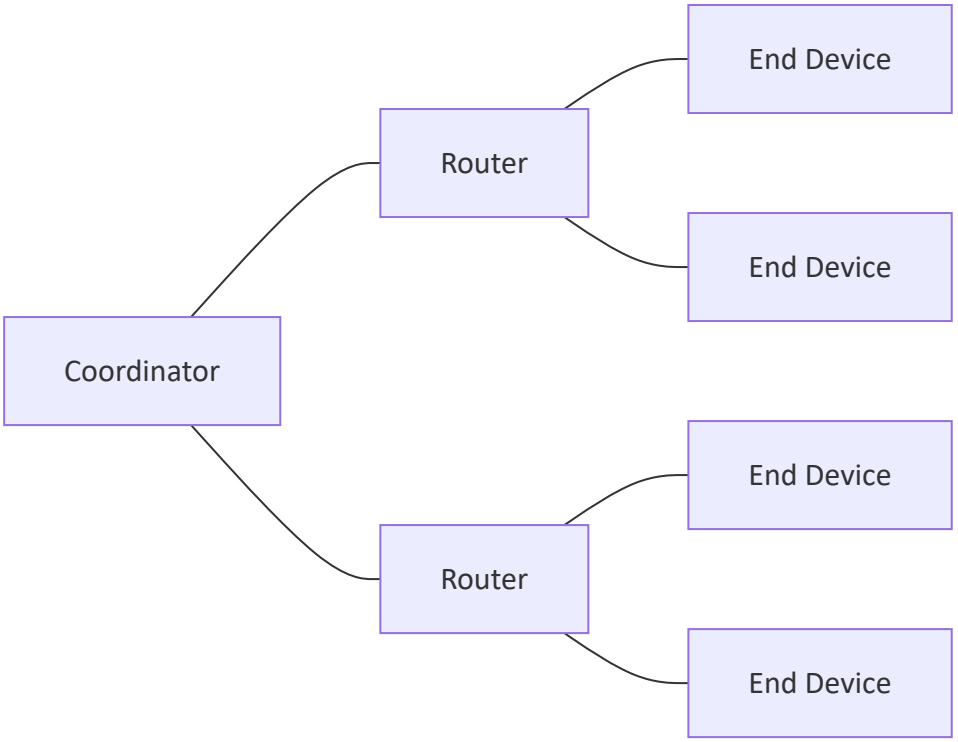
Operational Modes:

Active Mode:

- Device is fully operational.
- Transmits and receives data.
- Higher power consumption.

Sleep Mode:

- Device conserves power.
- Wakes periodically to check.
- Lower power consumption.
- Used by end devices.



Q16. Explain the core IoT functional stack, including its layers and the role of each layer.

Core IoT Functional Stack:

- Fundamental architecture of IoT.
- Defines essential layers and functions.
- Enables standardized IoT development.
- Covers device to application continuum.

Things Layer:

- Physical devices and sensors.
- Actuators and controllers.
- Data generation and action.

Communication Layer:

- Network connectivity.
- Protocol implementation.
- Data transmission.

Data Processing Layer:

- Data aggregation and filtering.
- Analytics and transformation.
- Edge and cloud processing.

Application Layer:

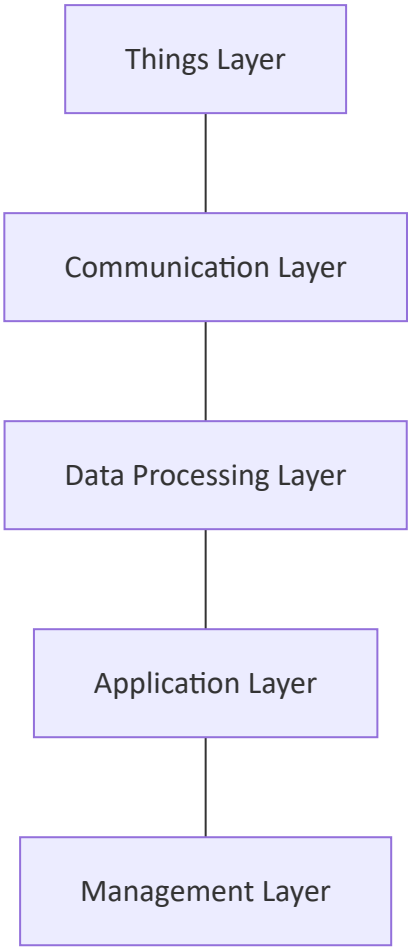
- User-facing applications.
- Business logic implementation.
- APIs and dashboards.

Management Layer:

- Device management.
- Network management.
- Security management.

Role of Each Layer:

Layer	Role
Things	Generate data and perform actions
Communication	Transmit data reliably
Data Processing	Analyze and transform data
Application	Deliver services to users
Management	Monitor and control system



2 Marks Questions

Q2. Explain Li-Fi technology and its applications in the context of IoT.

Li-Fi:

- Light Fidelity uses visible light communication.
- Transmits data using LED light.
- High speed compared to Wi-Fi.

- Cannot penetrate walls, secure transmission.

Applications in IoT:

- Indoor positioning systems.
- Smart lighting and home automation.
- Hospital and aircraft communication.
- Underwater IoT applications.

Q4. Define smart objects, and explain their characteristics and role in IoT.

Smart Objects:

- Physical objects with computing capabilities.
- Connected to network for communication.
- Sense and react to environment.

Characteristics:

- Unique identification.
- Sensing and actuation ability.
- Network connectivity.
- Local processing capability.
- Energy efficiency.

Role in IoT:

- Generate data from physical world.
- Act as building blocks of IoT.
- Enable automation and control.

Q7. Explain Z-Wave technology, its working principle, and its applications in IoT.

Z-Wave:

- Wireless mesh network protocol.
- Operates at 908 MHz frequency.
- Low power and low data rate.
- Designed for home automation.

Working Principle:

- Controller manages network.
- Devices form mesh network.
- Messages routed through nodes.
- Range about 30 meters per hop.

Applications:

- Smart home lighting control.
- Security systems and locks.
- Thermostats and HVAC control.

Q13. Differentiate between data analytics and network analytics in the context of IoT.

Factor	Data Analytics	Network Analytics
Focus	Data content and patterns	Network traffic and performance
Purpose	Extract business insights	Monitor network health
Input	Sensor data and events	Network packets and logs
Output	Predictions and reports	Performance metrics
Tools	ML, statistics	Network monitoring tools
Users	Data scientists	Network administrators

Q14. Differentiate between analytics and control applications in IoT systems.

Factor	Analytics Applications	Control Applications
Purpose	Analyze data for insights	Control physical processes
Timing	Batch or real-time	Real-time deterministic
Output	Reports and predictions	Commands and actions
Latency	Moderate acceptable	Very low required
Examples	Predictive maintenance	Smart thermostat control
Criticality	Business decisions	Safety and operations

Q15. Differentiate between data analytics and network analytics, and explain how data analytics contributes to business benefits.

Factor	Data Analytics	Network Analytics
Focus	Data content and meaning	Network traffic patterns
Purpose	Business insights	Network optimization
Input	Sensor and business data	Packets and flow data
Output	Predictions and reports	Performance metrics

Business Benefits of Data Analytics:

- Improves decision making with insights.
- Reduces operational costs.
- Enables predictive maintenance.
- Increases customer satisfaction.
- Identifies new revenue opportunities.